RAJALAKSHMI ENGINEERING COLLEGE

Dhanusheya G 251801053-2025-AIDS C2

Started on	Thursday, 27 August 2026, 8:52 AM
State	Finished
Completed on	Thursday, 17 September 2026, 9:41 AM
Time taken	21 days
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as “No”.

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value “x”)

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

// Recursive Binary Search function to find a target value in a sorted array range [low, high]
int binarySearch(int arr[], int low, int high, int target) {
    if (low > high) {
        return -1; // Not found
    }

    int mid = low + (high - low) / 2;

    if (arr[mid] == target) {
        return mid; // Found target index
    } else if (arr[mid] > target) {
        return binarySearch(arr, low, mid - 1, target); // Search left half
    } else {
        return binarySearch(arr, mid + 1, high, target); // Search right half
    }
}

// Divide and Conquer wrapper function to check for the two elements
int findTwoElements(int arr[], int low, int high, int x, int *elem1, int *elem2) {
    // Base case: if low crosses high, no pair found in this branch
    if (low >= high) {
        return 0;
    }

    // Divide: Pick the current element as the first candidate
    int i = low;
    int target = x - arr[i];

    // Conquer: Use recursive binary search to find the complement in the rest of the array [i + 1, high]
    int j = binarySearch(arr, i + 1, high, target);

    if (j != -1 && i != j) {
        *elem1 = arr[i];
        *elem2 = arr[j];
        return 1; // Pair found
    }

    // Otherwise, recursively check the rest of the array (Divide and Conquer step)
    return findTwoElements(arr, low + 1, high, x, elem1, elem2);
}

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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